Fiscal Year:	FY 2024	**Task Last Updated:**	FY 07/16/2024
PI Name:	Burke, Shawn Ph.D.		
Project Title:	Facilitating the Synergistic Side of Cultural Differences in LDSE: Identification of Challenges and Development of Cultural Training		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Behavior and performance		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	(1) **HSIA**:Risk of Adverse Outcomes Due to Inadequate Human Systems Integration Architecture (2) **Team**:Risk of Performance and Behavioral Health Decrements Due to Inadequate Cooperation, Coordination, Communication, and Psychosocial Adaptation within a Team		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2014-15 HERO NNJ14ZSA001N-MIXEDTOPICS. Appendix E: Behavioral Health & Human Health Countermeasures Topics
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No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA JSC
Contact Monitor:	Whitmire, Alexandra	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 7/31/2025 per L.Dang/HFBP (Ed., 4/14/25) NOTE: End date changed to 5/31/2025 per D. Kulkarni/HFBP (Ed., 7/12/24) NOTE: End date changed to 8/21/2024 per NSSC information (Ed., 8/23/23) NOTE: End date changed to 8/21/2023 per NSSC information (Ed., 7/22/22) NOTE: End date changed to 8/21/2022 per NSSC information (Ed., 8/24/21) NOTE: End date changed to 8/21/2021 per NSSC information (Ed., 7/27/20)		
Key Personnel Changes/Previous PI:	July 2023 report: No key personnel changes		

COI Name (Institution):	Salas, Eduardo Ph.D. (Rice University)
Grant/Contract No.:	NNX16AP96G
Performance Goal No.:	
Performance Goal Text:	
Task Description:	Culture has been a factor for space exploration since the mid 1970s when the U.S. first partnered with Russia. Work has indicated that culturally-based differences in values, beliefs, and preferences for cognition and action can have an impact on interdependent action. Cultural differences can pose challenges for a number of transition, action, and interpersonal processes and emergent states in teams. However, almost all of this work has been conducted outside the context of spaceflight. The little work that has been conducted within spaceflight has begun to suggest that the impact of culture on teams may be different in long-duration space exploration (LDSE). This drives the question as to whether cultural differences within spaceflight crews has the same impact on team performance as that reported within the wider literature on cross-cultural teams and what can be done to create synergy. We leverage prior work on cultural differences and teams in mission critical environments and archival data on team interaction in ICE (isolated, confined environments) and pair this with experimentation in ground analog studies to better understand the critical challenges created by cultural differences in LDSE and use this information to create and test cultural training tools that can be used pre- and in-flight. Specifically, we take a multi-pronged approach to answer the following questions: (1) what are the critical issues surround culture and team performance in LDSE, (2) from a compositional standpoint is there an 'ideal' team profile in terms of cultural orientations to facilitate team performance and adaptability in LDSE, (3) what types of cultural training can be developed for use pre-flight and in-flight, and (4) what tools can be developed to facilitate practitioners in maximizing the potential synergy in culturally different teams operating in LDSE (e.g., pillars, guidelines, practices, tips).
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Within this project, we seek to answer the following questions: (1) what are the critical issues surrounding culture and team performance in long-duration spaceflight, (2) is there a team cultural profile that facilitates team performance and adaptation in long-duration spaceflight, and (3) what tools can be developed to facilitate synergy/mitigate decrements in teamwork and team performance within culturally different teams operating in the context of spaceflight. The proposed project will impact numerous areas. First, this research will address current theoretical gaps surrounding the critical challenge of cultural differences in long-duration spaceflight (both on the ground and in orbit). This information is then used to build a series of complementary, modular training tools to facilitate the crew's ability to turn cultural differences into a synergistic force. The existent literature on cross-cultural training does little to focus on cultural differences in crews operating in isolated, confined environments. Additionally, the focus of most cross-cultural training programs is to prepare expatriates to go abroad; therefore, teaching cross-cultural competence with regard to teamwork skills is rarely seen. In building our training we seek to validate the effectiveness of such programs within analog environments using sound scientific methods and training evaluation techniques. Finally, as an important part of training development is the assessment of the skills gained during training we will also deliver a set of assessment tools that can be used to diagnose cross-cultural competence levels with regard to team interaction.
Task Progress:	During this reporting period the primary activities have focused upon the following thrusts: (1) data integration and analysis using a 4- and 8-month analog mission, with a specific focus on the impact of cultural values on crew dynamics, (2) development of cultural training, (3) preparation and participation in the Human Exploration Research Analog (HERA) campaign 7 – Missions 1 and 2, and (4) initial work on a grant augmentation award to examine the effect of cultural differences on shared mental models and the effect of the interaction on affect and its regulation. With respect to the 4 and 8-month analog missions, we began to run comparative analyses between the two missions to examine what trends remained consistent across the two missions in terms of how cultural values impact crew dynamics as well as what may seem more unique to particular mission parameters. In doing so, we also examined the role that temporal factors may occupy in the variability of relationships. Within the past reporting period, we now have all the data from the 4- and 8-month missions, including that which was gained through data-sharing agreements. Therefore, we have begun to extend our comparative analyses across the broader set of constructs – not only focusing on the role that cultural values occupy in crew dynamics in terms of team process, but how they may relate to crew affect as well as crew cognition (through the lens of mental models). This past reporting period we have also been examining the journal data provided by crew members to not only augment the survey data but also to provide insight into nuances seen in temporal differences across the two missions in the strength of key relationships. The journal data provides a rich source of qualitative data currently representing approximately 608 journal entries across twelve individuals comprising two crews. Journals were examined as a set using a qualitative exploratory approach. The obtained data were thematically coded using Atlas.ti software. For the thematic analysis, two coders coded the journals independently for cultural differences and team dynamics-related aspects. Throughout the coding procedure, agreement meetings were periodically held to ensure consistency in the coding scheme. Findings were extracted based on the resulting data structure and emerging patterns. In addition to integrating existing data and conducting analyses on that integrated set, both quantitative and qualitative we have also prepared to participate in a shorter analog mission where we will conduct an initial test of training that was developed this past reporting period. The training program consists of five modules and reflects a combination of knowledge-based information and practical guidance based on results from data collection in prior years. It also incorporates a module which dives deeper into one specific strategy that the crew can utilize both pre- and in-mission to "stay on the same page" and assists the team in creating that third culture. It also provides them with hands-on practice and discussion opportunities to begin to build a set of norms, practices, and understanding that is specifically attuned to the fact that the crew is multicultural. We are in the initial stages of testing the training and gathering feedback concerning those aspects crews find most useful and what additions they would find useful. Finally, we continue to collect our standard set of measures that were collected in the 4- and 8-month missions. At this point, two of four missions within the shorter analog have been completed. Moving forward we will continue to explore the nuances of the journal data to gain additional insight into the role of temporal factors in explaining fluctuations in

	what was salient to each crew as well as helping to explain fluctuations in key relationships across time. Furthermore, we will integrate the results of the ongoing data collection. Again, exploring what findings hold and what may seem to vary. We will continue to utilize and integrate the information gained into the refinement of the products we are building.
Bibliography Type:	Description: (Last Updated: 07/15/2026)
Books/Book Chapters	Shorey A, Moran LH, Wiese CW, Burke CS. "From performance to well-being: Broadening the team resilience discourse." in "Research in Occupational Stress and Well-Being: Stress and Well-Being in Teams." Ed. P.D. Harms, D. Chang. UK: Emerald Publishing, in press, Oct-2024
Books/Book Chapters	Dinh JV, Salas E. "Culture and learning, training, and development." in "In: The Oxford Handbook of Cross-Cultural Organizational Behavior." New York: Oxford University Press, 2024. p. 110., Jan-2024